ASSIGNMENT-11.3

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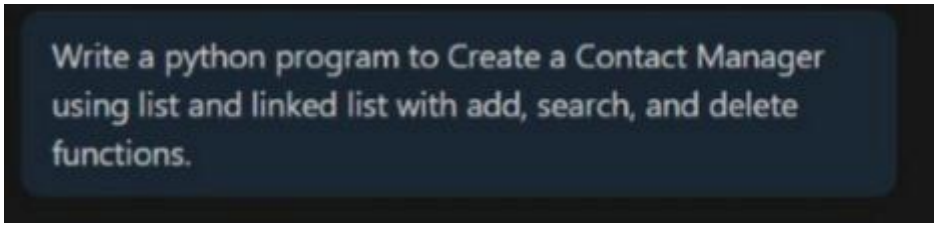
Batch: 05

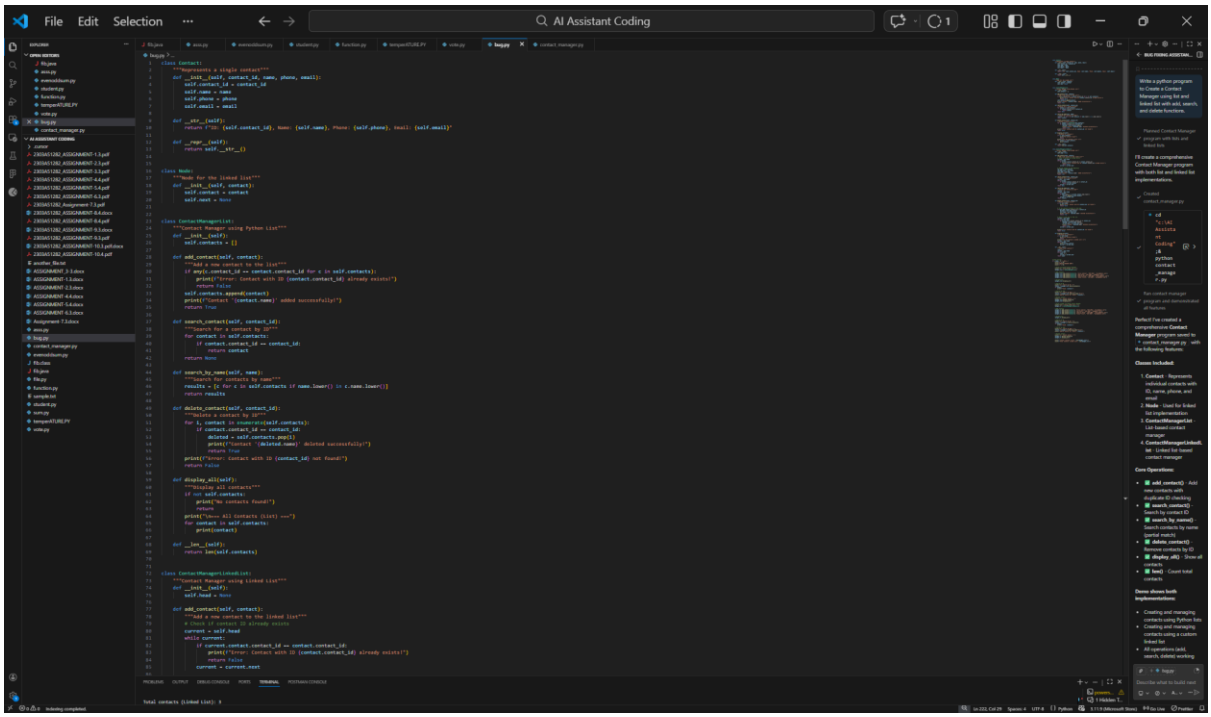
Task 1: Smart Contact Manager (Arrays & Linked Lists)
Scenario
SR University's student club requires a simple Contact Manager Application to store members' names and phone numbers. The system should support efficient addition, searching, and deletion of contacts.

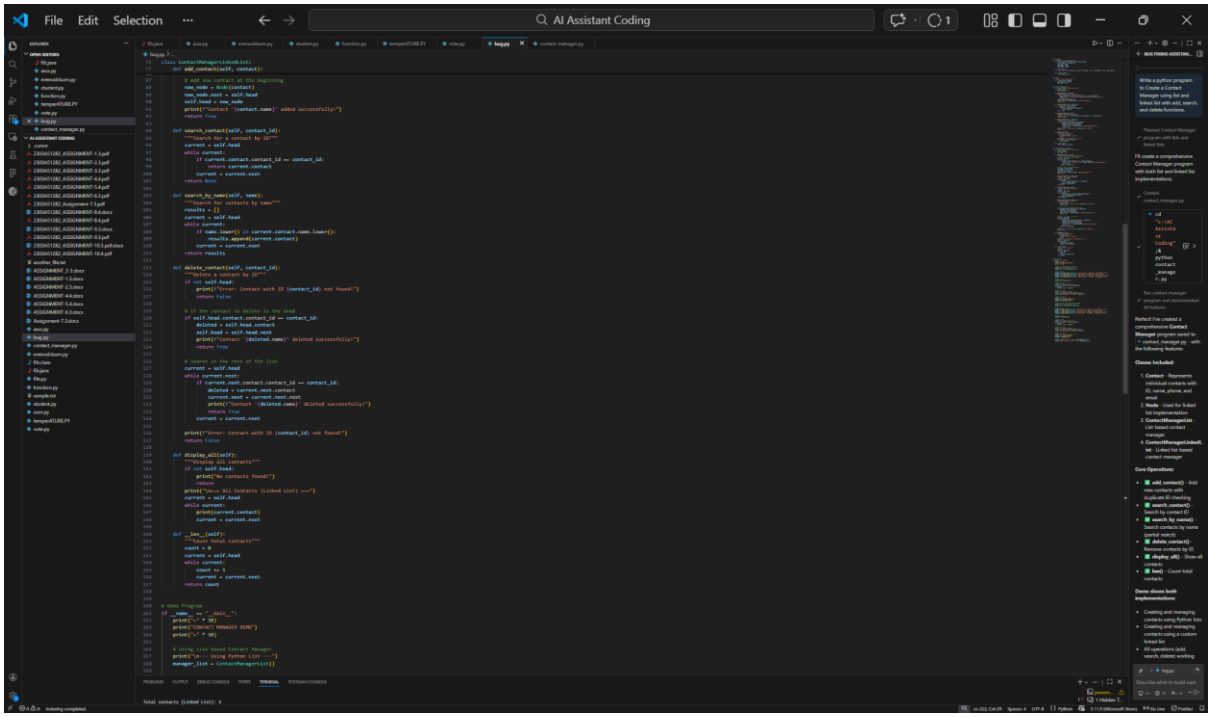
Prompt:

Write a python program to Create a Contact Manager using list and linked list with add, search, and delete functions.



Code:


File Edit Selection
AI Assistant Coding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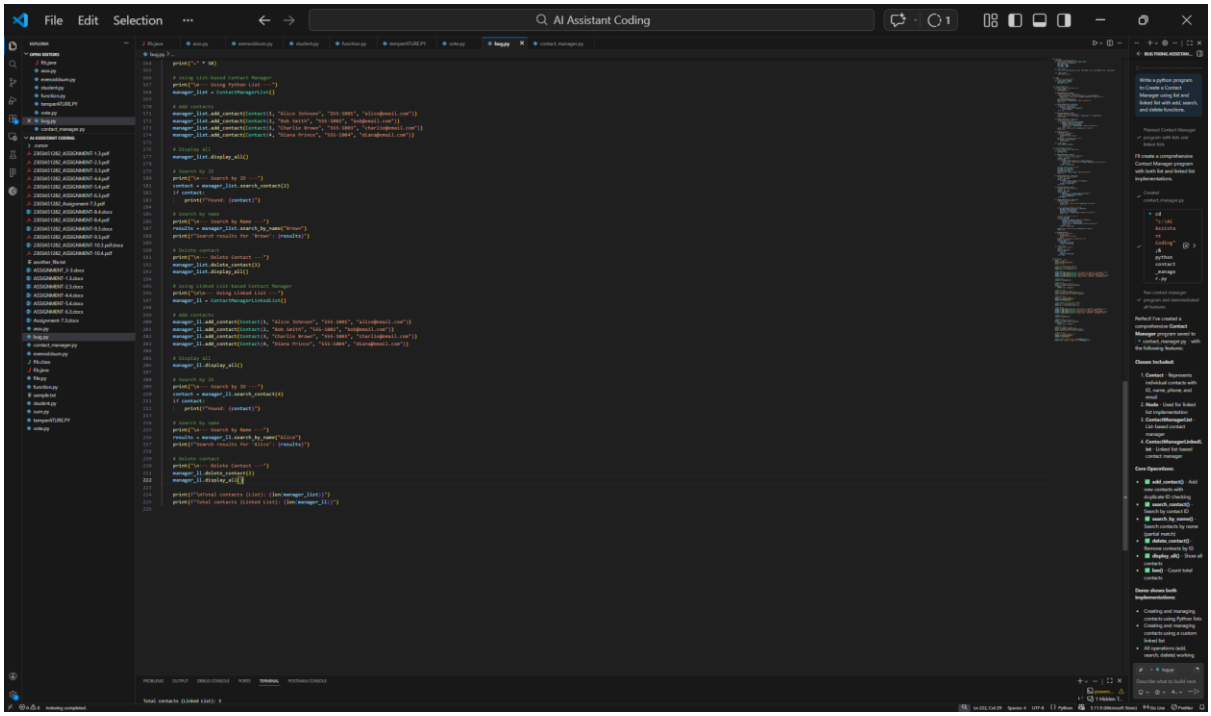
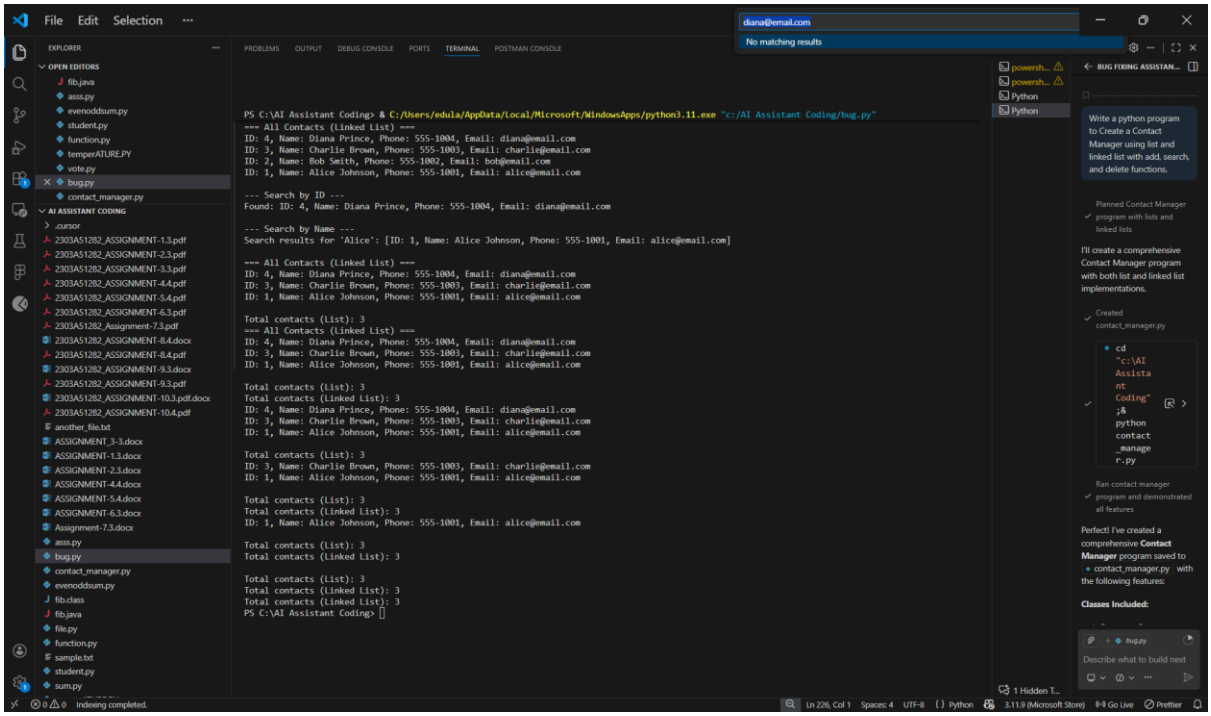
AI Assistant Coding



Output:



Explanation:

- In an array, adding at the end is fast, but inserting in the middle is slow because elements must shift.
- In a linked list, insertion is fast because no shifting is needed.
- Searching takes the same time in both (you must check each element).
- Deleting in an array is slower due to shifting elements.
- Linked list is better for frequent insertions and deletions.

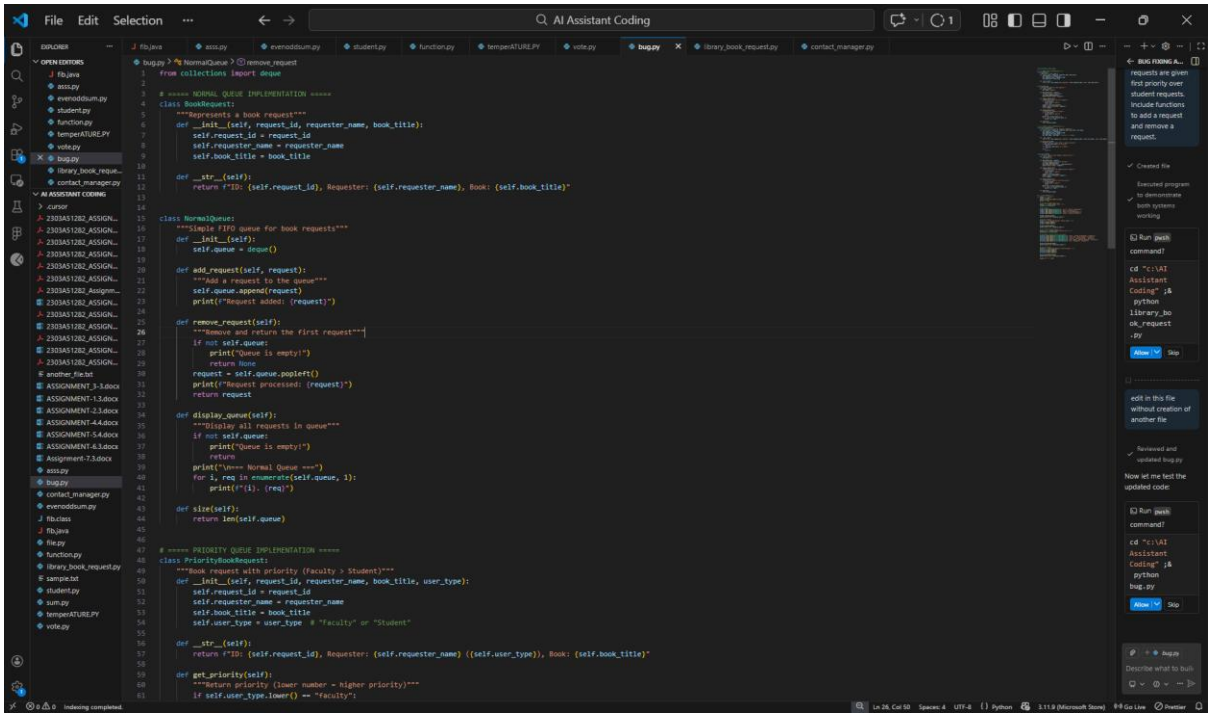
Task 2: Library Book Search System (Queues & Priority Queues)
Scenario
The SRU Library manages book borrow requests. Students and faculty submit requests, but faculty requests must be prioritized over student requests.

Prompt:

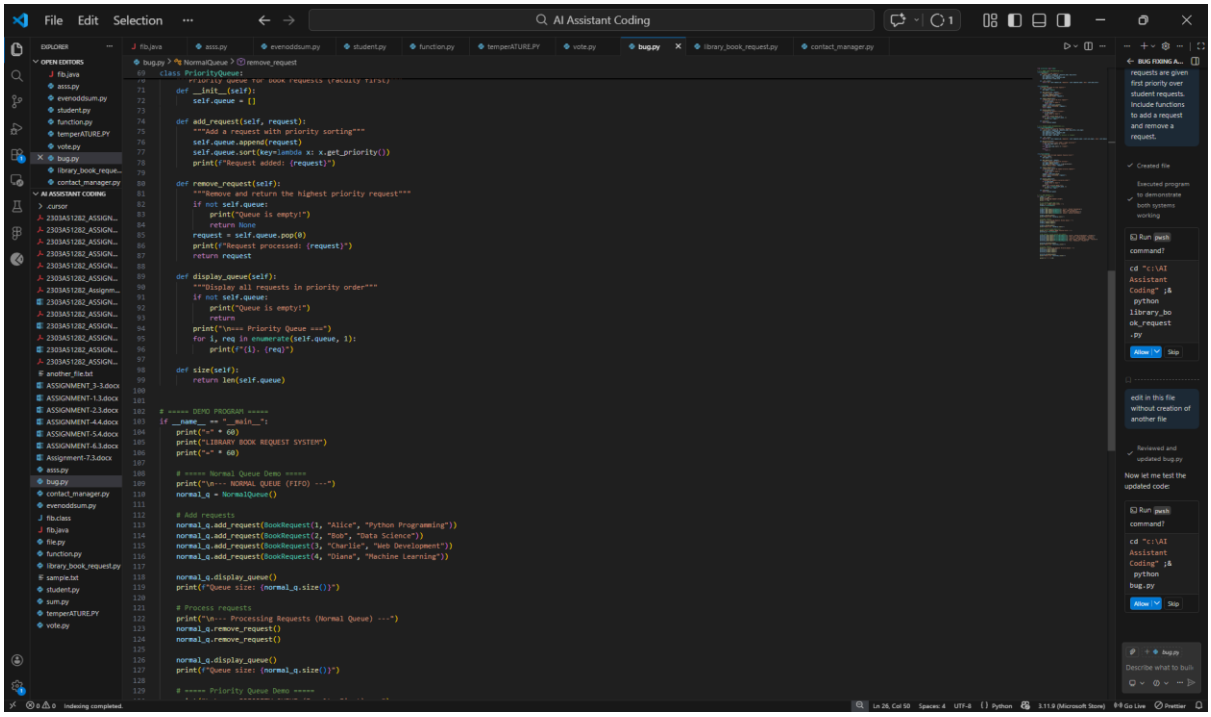
Write a Python program for a library book request system.First, make a normal queue where requests are handled in the order they come.Then, make another version where faculty requests are given first priority over student requests.Include functions to add a request and remove a request.

Write a Python program for a library book request system.
First, make a normal queue where requests are handled in the order they come.
Then, make another version where faculty requests are given first priority over student requests.
Include functions to add a request and remove a request.

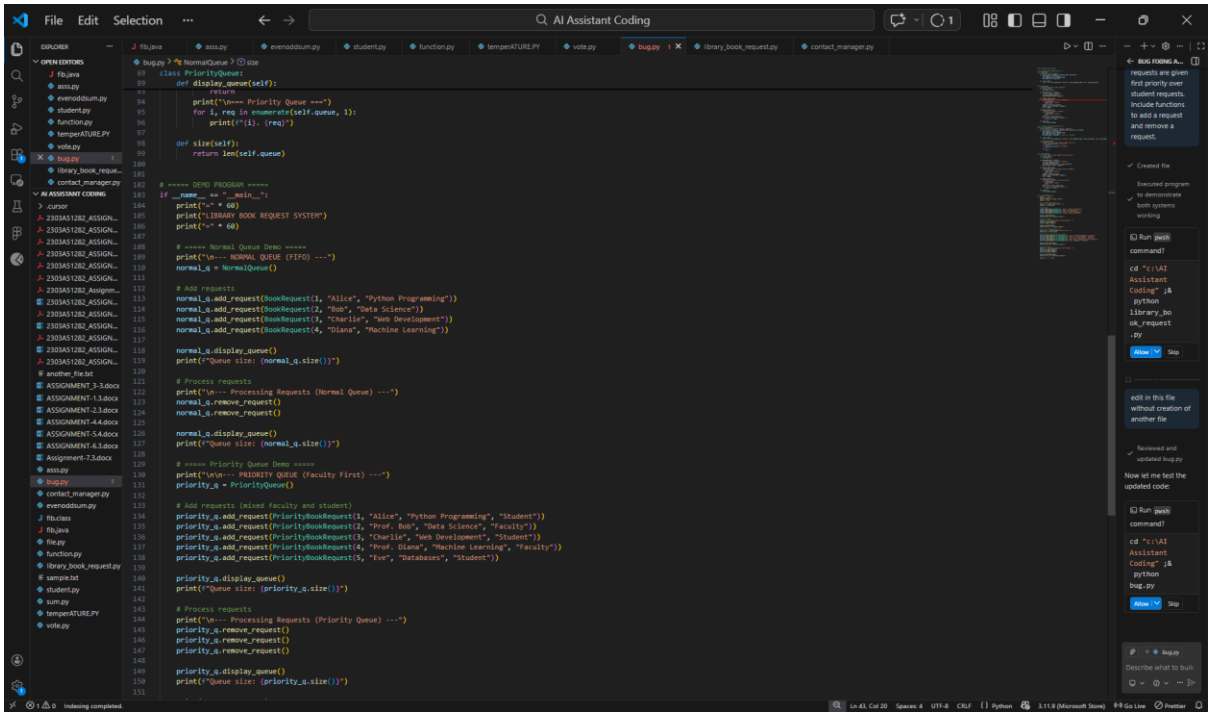
Code:



```python
from collections import deque

# ===== NORMAL QUEUE IMPLEMENTATION =====
class BookRequest:
    """Represents a book request"""
    def __init__(self, request_id, requester_name, book_title):
        self.request_id = request_id
        self.requester_name = requester_name
        self.book_title = book_title

    def __str__(self):
        return f"ID: {self.request_id}, Requester: {self.requester_name}, Book: {self.book_title}"


class NormalQueue:
    """Simple FIFO queue for book requests"""
    def __init__(self):
        self.queue = deque()

    def add_request(self, request):
        """Add a request to the queue"""
        self.queue.append(request)
        print(f"Request added: {request}")

    def remove_request(self):
        """Remove and return the first request"""
        if not self.queue:
            print("Queue is empty!")
            return None
        request = self.queue.popleft()
        print(f"Request processed: {request}")
        return request

    def display_queue(self):
        """Display all requests in queue"""
        if not self.queue:
            print("Queue is empty!")
            return
        print("\n=== Normal Queue ===")
        for i, req in enumerate(self.queue, 1):
            print(f"{i}. {req}")

    def size(self):
        return len(self.queue)


# ===== PRIORITY QUEUE IMPLEMENTATION =====
class PriorityBookRequest:
    """Book request with priority (Faculty > Student)"""
    def __init__(self, request_id, requester_name, book_title, user_type):
        self.request_id = request_id
        self.requester_name = requester_name
        self.book_title = book_title
        self.user_type = user_type  # "Faculty" or "Student"

    def __str__(self):
        return f"ID: {self.request_id}, Requester: {self.requester_name} ({self.user_type}), Book: {self.book_title}"

    def get_priority(self):
        """Return priority (lower number = higher priority)"""
        if self.user_type.lower() == "faculty":
```



```python
class PriorityQueue:
    """Priority queue for book requests (faculty first)"""
    def __init__(self):
        self.queue = []

    def add_request(self, request):
        """Add a request with priority sorting"""
        self.queue.append(request)
        self.queue.sort(key=lambda x: x.get_priority())
        print(f"Request added: {request}")

    def remove_request(self):
        """Remove and return the highest priority request"""
        if not self.queue:
            print("Queue is empty!")
            return None
        request = self.queue.pop(0)
        print(f"Request processed: {request}")
        return request

    def display_queue(self):
        """Display all requests in priority order"""
        if not self.queue:
            print("Queue is empty!")
            return
        print("\n=== Priority Queue ===")
        for i, req in enumerate(self.queue, 1):
            print(f"{i}. {req}")

    def size(self):
        return len(self.queue)


# ===== DEMO PROGRAM =====
if __name__ == "__main__":
    print("=" * 60)
    print("LIBRARY BOOK REQUEST SYSTEM")
    print("=" * 60)

    # ===== Normal Queue Demo =====
    print("\n--- NORMAL QUEUE (FIFO) ---")
    normal_q = NormalQueue()

    # Add requests
    normal_q.add_request(BookRequest(1, "Alice", "Python Programming"))
    normal_q.add_request(BookRequest(2, "Bob", "Data Science"))
    normal_q.add_request(BookRequest(3, "Charlie", "Web Development"))
    normal_q.add_request(BookRequest(4, "Diana", "Machine Learning"))

    normal_q.display_queue()
    print(f"Queue size: {normal_q.size()}")

    # Process requests
    print("\n--- Processing Requests (Normal Queue) ---")
    normal_q.remove_request()
    normal_q.remove_request()

    normal_q.display_queue()
    print(f"Queue size: {normal_q.size()}")

    # ===== Priority Queue Demo =====
```



```python
class PriorityQueue:
    def display_queue(self):
        return
        print("\n=== Priority Queue ===")
        for i, req in enumerate(self.queue, 1):
            print(f"{i}. {req}")

    def size(self):
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    normal_q.display_queue()
    print(f"Queue size: {normal_q.size()}")

    # Process requests
    print("\n--- Processing Requests (Normal Queue) ---")
    normal_q.remove_request()
    normal_q.remove_request()

    normal_q.display_queue()
    print(f"Queue size: {normal_q.size()}")

    # ===== Priority Queue Demo =====
    print("\n\n--- PRIORITY QUEUE (Faculty First) ---")
    priority_q = PriorityQueue()

    # Add requests (mixed faculty and student)
    priority_q.add_request(PriorityBookRequest(1, "Alice", "Python Programming", "Student"))
    priority_q.add_request(PriorityBookRequest(2, "Prof. Bob", "Data Science", "Faculty"))
    priority_q.add_request(PriorityBookRequest(3, "Charlie", "Web Development", "Student"))
    priority_q.add_request(PriorityBookRequest(4, "Prof. Diana", "Machine Learning", "Faculty"))
    priority_q.add_request(PriorityBookRequest(5, "Eve", "Databases", "Student"))

    priority_q.display_queue()
    print(f"Queue size: {priority_q.size()}")

    # Process requests
    print("\n--- Processing Requests (Priority Queue) ---")
    priority_q.remove_request()
    priority_q.remove_request()
    priority_q.remove_request()

    priority_q.display_queue()
    print(f"Queue size: {priority_q.size()}")
```
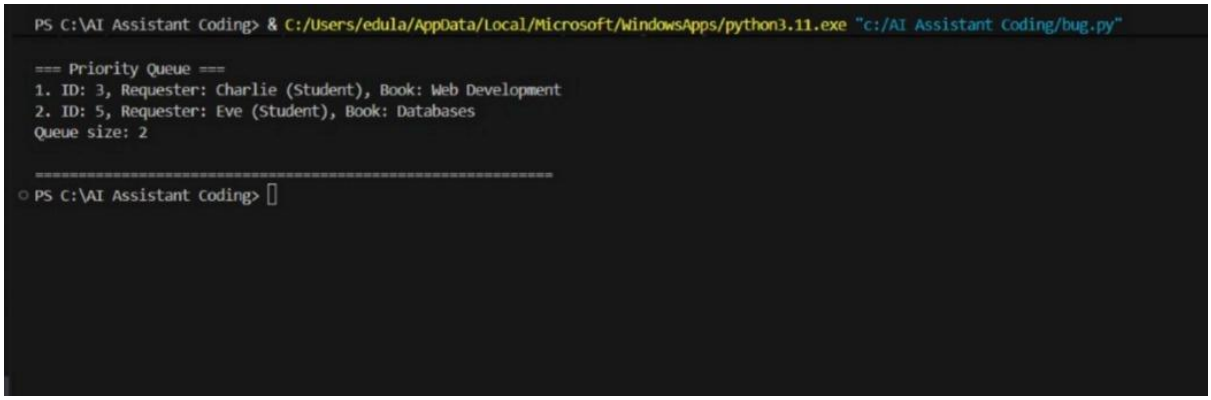

Output:



```
PS C:\AI Assistant Coding> & C:/Users/edula/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/AI Assistant Coding/bug.py"

=== Priority Queue ===
1. ID: 3, Requester: Charlie (Student), Book: Web Development
2. ID: 5, Requester: Eve (Student), Book: Databases
Queue size: 2

============================================================
PS C:\AI Assistant Coding>
```

Explanation:

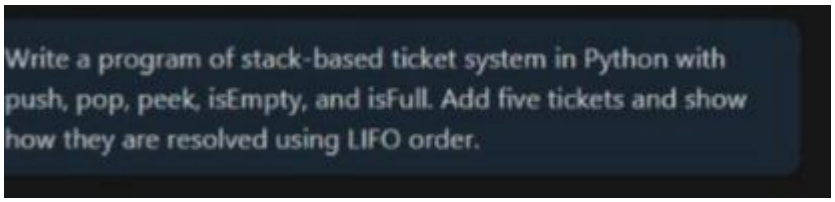
- Queue (FIFO) → First request comes, first served.(If a student requests first, they get the book first.)
- Priority Queue → Faculty requests are served before students, even if they come later.
- enqueue() → Adds a request to the system.

- dequeue() → Removes and processes the next request.

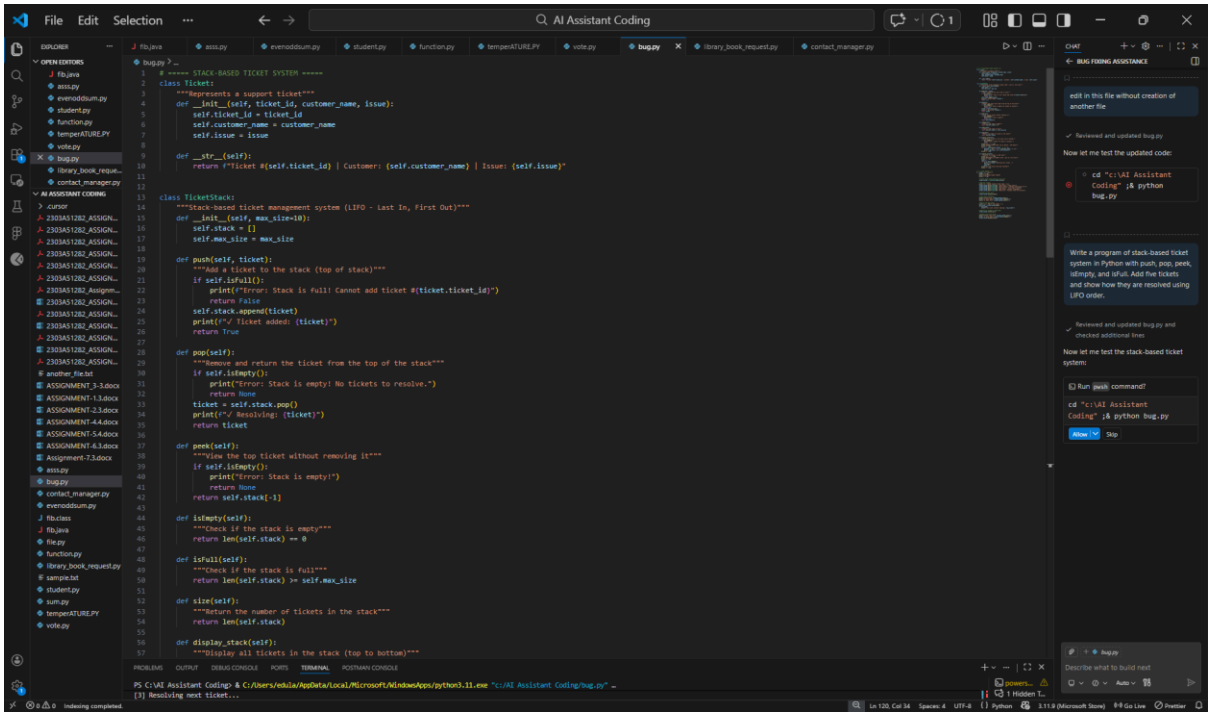
Task 3: Emergency Help Desk (Stack Implementation)
Scenario
SR University's IT Help Desk receives technical support tickets from students and staff. While tickets are received sequentially, issue escalation follows a Last-In, First-Out (LIFO) approach.

Prompt:

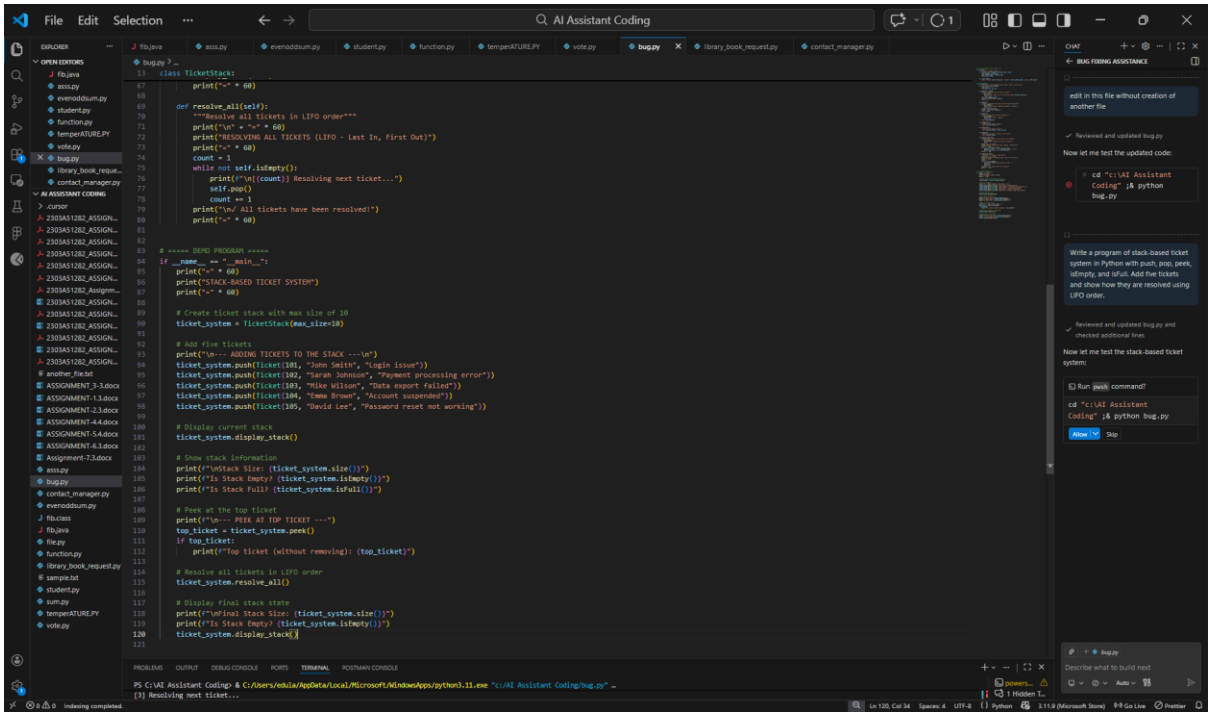
Write a program of stack-based ticket system in Python with push, pop, peek, isEmpty, and isFull. Add five tickets and show how they are resolved using LIFO order.



Code:

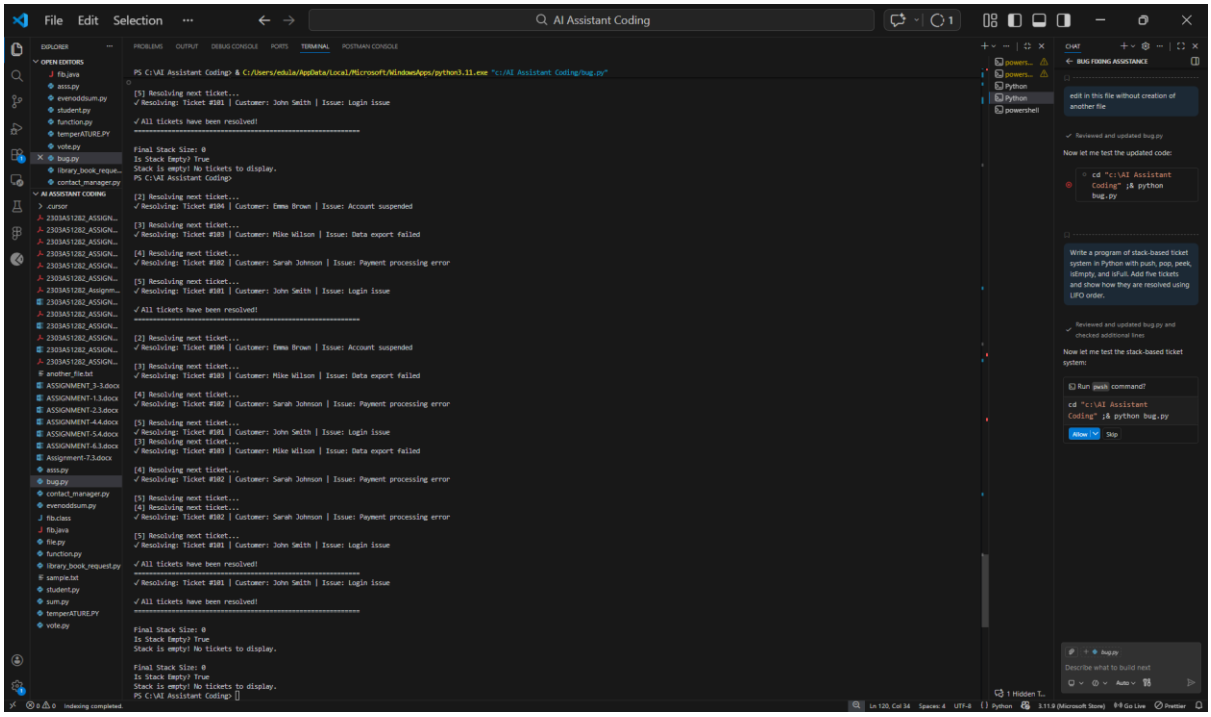


```python
# ===== STACK-BASED TICKET SYSTEM =====
class Ticket:
    """Represents a support ticket"""
    def __init__(self, ticket_id, customer_name, issue):
        self.ticket_id = ticket_id
        self.customer_name = customer_name
        self.issue = issue

    def __str__(self):
        return f"Ticket #{self.ticket_id} | Customer: {self.customer_name} | Issue: {self.issue}"

class TicketStack:
    """Stack-based ticket management system (LIFO - Last In, First Out)"""
    def __init__(self, max_size=10):
        self.stack = []
        self.max_size = max_size

    def push(self, ticket):
        """Add a ticket to the stack (top of stack)"""
        if self.isFull():
            print("Error: Stack is full! Cannot add ticket #{ticket.ticket_id}")
            return False
        self.stack.append(ticket)
        print(f"✓ Ticket added: {ticket}")
        return True

    def pop(self):
        """Remove and return the ticket from the top of the stack"""
        if self.isEmpty():
            print("Error: Stack is empty! No tickets to resolve.")
            return None
        ticket = self.stack.pop()
        print(f"✓ Resolving: {ticket}")
        return ticket

    def peek(self):
        """View the top ticket without removing it"""
        if self.isEmpty():
            print("Error: Stack is empty!")
            return None
        return self.stack[-1]

    def isEmpty(self):
        """Check if the stack is empty"""
        return len(self.stack) == 0

    def isFull(self):
        """Check if the stack is full"""
        return len(self.stack) >= self.max_size

    def size(self):
        """Return the number of tickets in the stack"""
        return len(self.stack)

    def display_stack(self):
        """Display all tickets in the stack (top to bottom)"""
```



```python
class TicketStack:
        print("=" * 60)

    def resolve_all(self):
        """Resolve all tickets in LIFO order"""
        print("\n" + "=" * 60)
        print("RESOLVING ALL TICKETS (LIFO - Last In, First Out)")
        print("=" * 60)
        count = 1
        while not self.isEmpty():
            print("\n[{count}] Resolving next ticket...")
            self.pop()
            count += 1
        print("\n✓ All tickets have been resolved!")
        print("=" * 60)


# ===== DEMO PROGRAM =====
if __name__ == "__main__":
    print("=" * 60)
    print("STACK-BASED TICKET SYSTEM")
    print("=" * 60)

    # Create ticket stack with max size of 10
    ticket_system = TicketStack(max_size=10)

    # Add five tickets
    print("\n--- ADDING TICKETS TO THE STACK ---\n")
    ticket_system.push(Ticket(101, "John Smith", "Login issue"))
    ticket_system.push(Ticket(102, "Sarah Johnson", "Payment processing error"))
    ticket_system.push(Ticket(103, "Mike Wilson", "Data export failed"))
    ticket_system.push(Ticket(104, "Emma Brown", "Account suspended"))
    ticket_system.push(Ticket(105, "David Lee", "Password reset not working"))

    # Display current stack
    ticket_system.display_stack()

    # Show stack information
    print(f"\nStack Size: {ticket_system.size()}")
    print(f"Is Stack Empty? {ticket_system.isEmpty()}")
    print(f"Is Stack Full? {ticket_system.isFull()}")

    # Peek at the top ticket
    print("\n--- PEEK AT TOP TICKET ---")
    top_ticket = ticket_system.peek()
    if top_ticket:
        print(f"Top ticket (without removing): {top_ticket}")

    # Resolve all tickets in LIFO order
    ticket_system.resolve_all()

    # Display final stack state
    print(f"\nFinal Stack Size: {ticket_system.size()}")
    print(f"Is Stack Empty? {ticket_system.isEmpty()}")
    ticket_system.display_stack()
```

Output:



Explanation:

The program uses a stack to manage help desk tickets.
A stack works in last in, first solved order.
When a new ticket is raised, it is added to the top.
When solving a ticket, the most recent one is handled first.
The program can also check if there are no tickets left or if the stack is full.

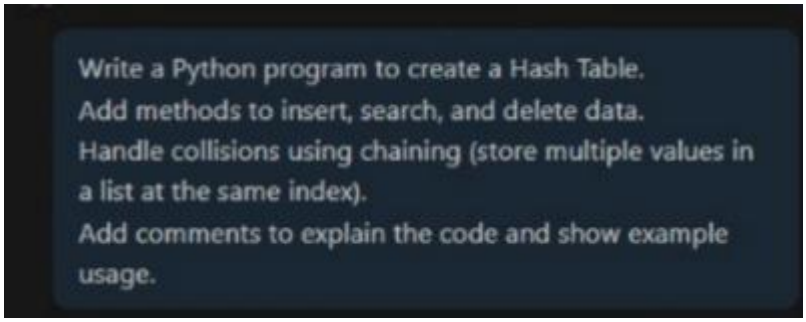
Task 4: Hash Table

Objective

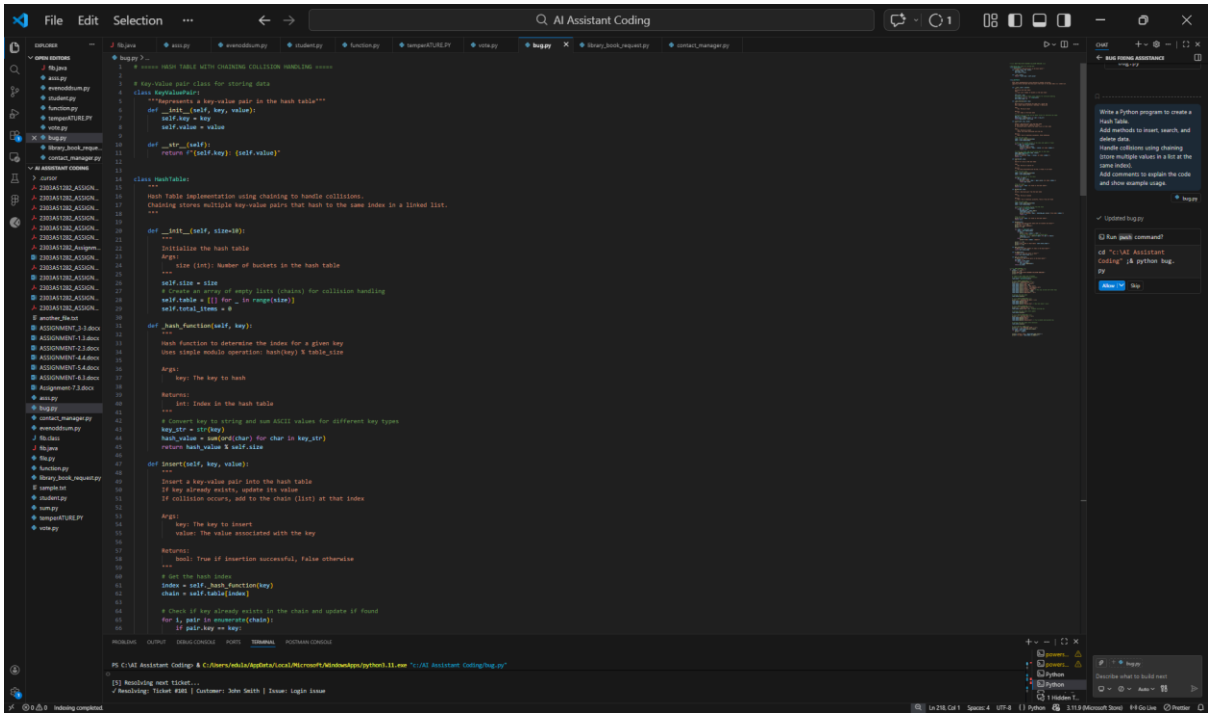
To implement a Hash Table and understand collision handling.

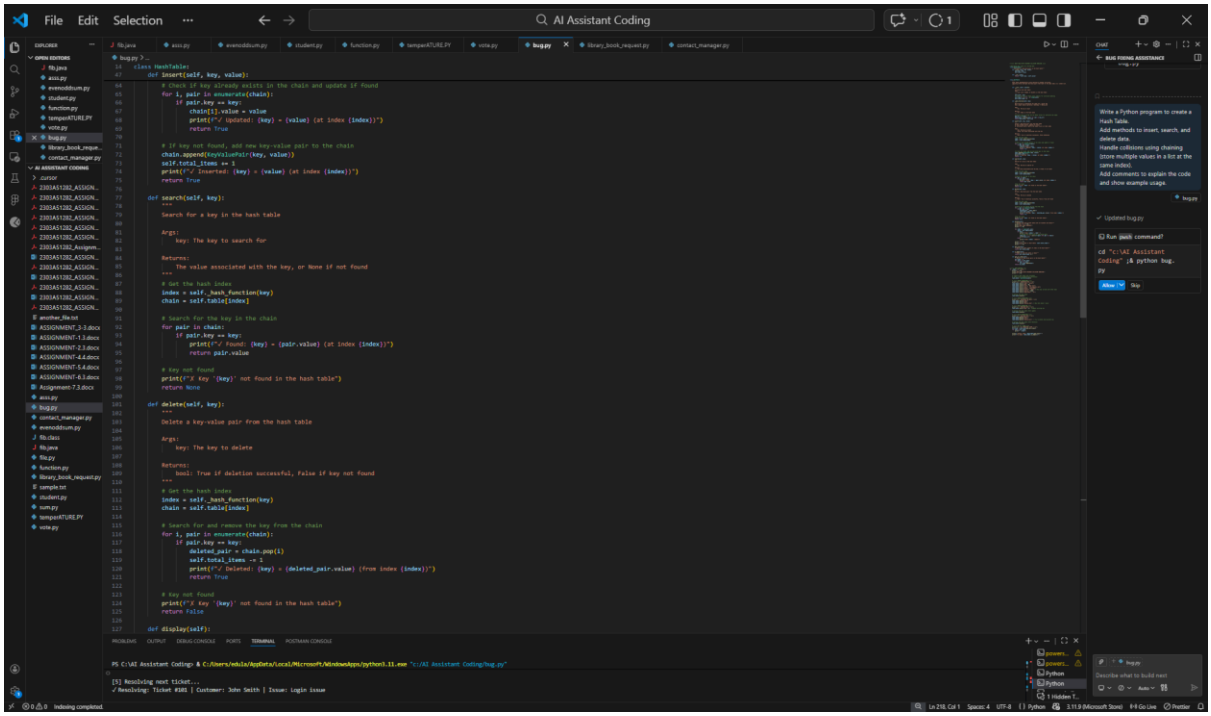
Prompt:

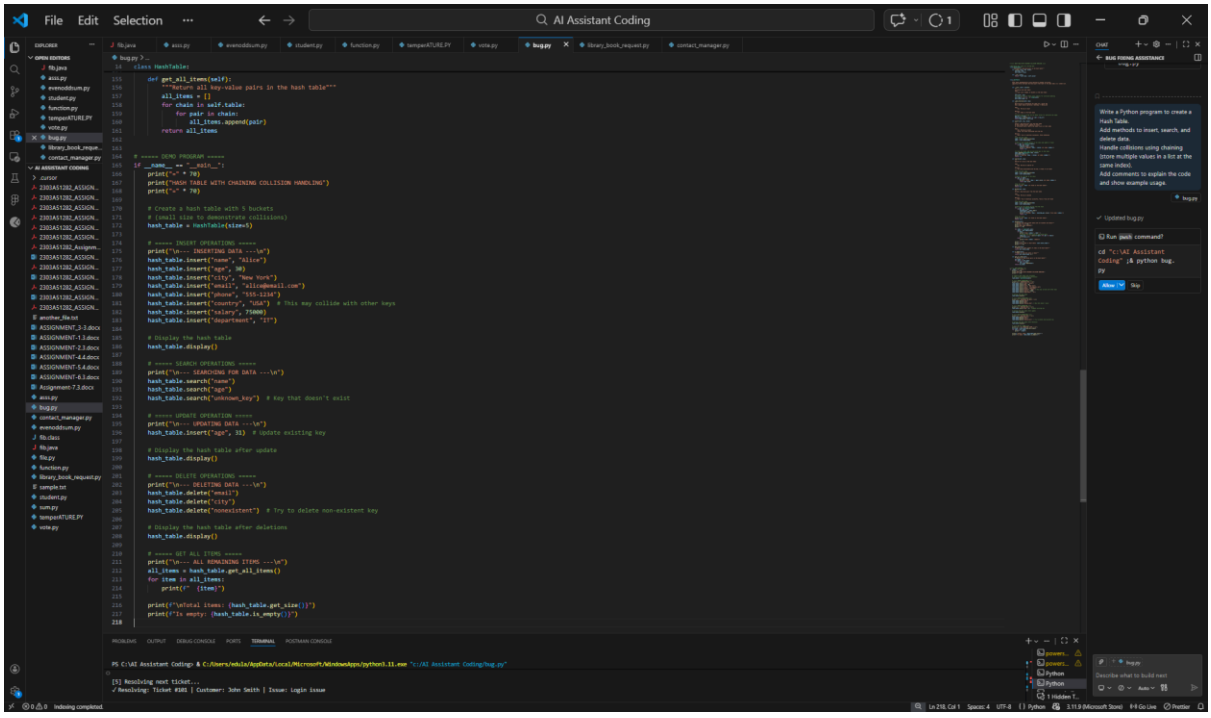
Write a Python program to create a Hash Table.
Add methods to insert, search, and delete data.
Handle collisions using chaining (store multiple values in a list at the same index).
Add comments to explain the code and show example usage.



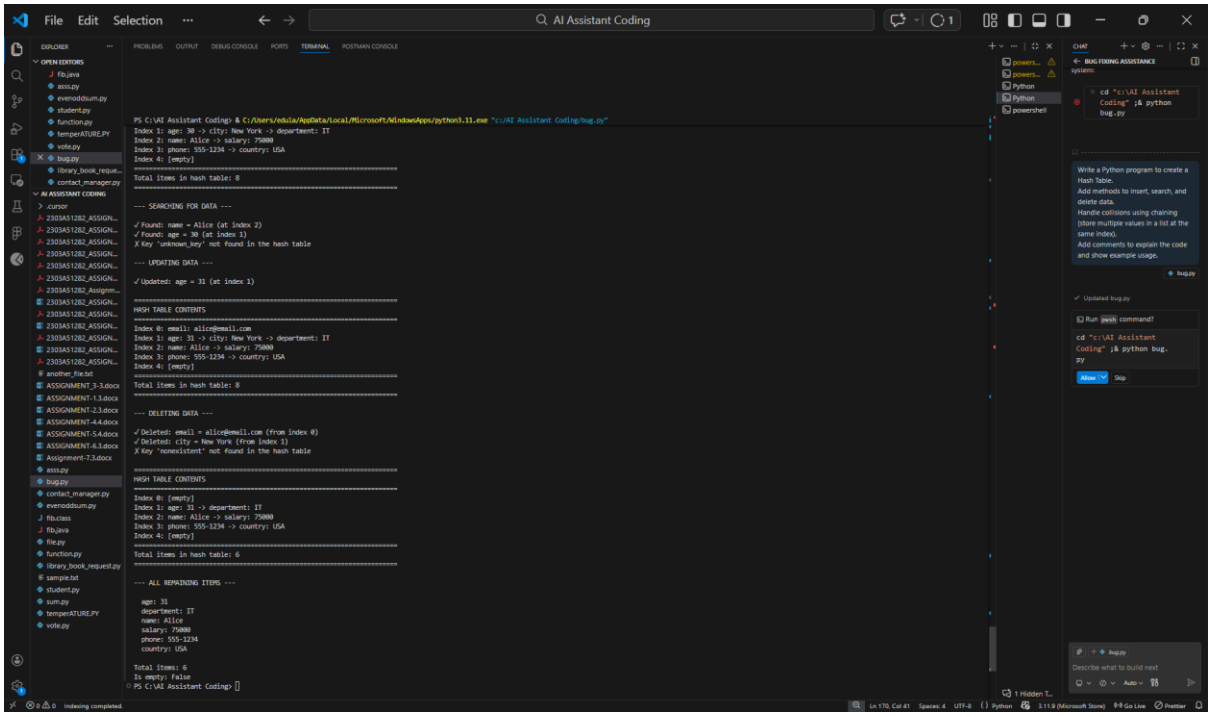
Code:




File Edit Selection
AI Assistant Coding
class HashTable:
def insert(self, key, value):
# Check if key already exists in the chain and update if found
for i, pair in enumerate(chain):
if pair.key == key:
chain[i].value = value
print(f"✓ Updated: {key} = {value} (at index {index})")
return True
# If key not found, add new key-value pair to the chain
chain.append(KeyValuePair(key, value))
self.total_items += 1
print(f"✓ Inserted: {key} = {value} (at index {index})")
return True
def search(self, key):
"""
Search for a key in the hash table
Args:
key: The key to search for
Returns:
The value associated with the key, or None if not found
"""
# Get the hash index
index = self._hash_function(key)
chain = self.table[index]
# Search for the key in the chain
for pair in chain:
if pair.key == key:
print(f"✓ Found: {key} = {pair.value} (at index {index})")
return pair.value
# Key not found
print(f"✗ Key '{key}' not found in the hash table")
return None
def delete(self, key):
"""
Delete a key-value pair from the hash table
Args:
key: The key to delete
Returns:
bool: True if deletion successful, False if key not found
"""
# Get the hash index
index = self._hash_function(key)
chain = self.table[index]
# Search for and remove the key from the chain
for i, pair in enumerate(chain):
if pair.key == key:
deleted_pair = chain.pop(i)
self.total_items -= 1
print(f"✓ Deleted: {key} = {deleted_pair.value} (from index {index})")
return True
# Key not found
print(f"✗ Key '{key}' not found in the hash table")
return False
def display(self):
PS C:\AI Assistant Coding> & C:/Users/mdula/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/AI Assistant Coding/bug.py"
[5] Resolving next ticket...
✓ Resolving: Ticket #101 | Customer: John Smith | Issue: Login issue


File Edit Selection
AI Assistant Coding
class HashTable:
def get_all_items(self):
"""Return all key-value pairs in the hash table"""
all_items = []
for chain in self.table:
for pair in chain:
all_items.append(pair)
return all_items
# ===== DEMO PROGRAM =====
if __name__ == "__main__":
print("=" * 70)
print("HASH TABLE WITH CHAINING COLLISION HANDLING")
print("=" * 70)
# Create a hash table with 5 buckets
# (small size to demonstrate collisions)
hash_table = HashTable(size=5)
# ===== INSERT OPERATIONS =====
print("\n--- INSERTING DATA ---\n")
hash_table.insert("name", "Alice")
hash_table.insert("age", 30)
hash_table.insert("city", "New York")
hash_table.insert("email", "alice@email.com")
hash_table.insert("phone", "555-1234")
hash_table.insert("country", "USA") # This may collide with other keys
hash_table.insert("salary", 75000)
hash_table.insert("department", "IT")
# Display the hash table
hash_table.display()
# ===== SEARCH OPERATIONS =====
print("\n--- SEARCHING FOR DATA ---\n")
hash_table.search("name")
hash_table.search("age")
hash_table.search("unknown_key") # Key that doesn't exist
# ===== UPDATE OPERATION =====
print("\n--- UPDATING DATA ---\n")
hash_table.insert("age", 31) # Update existing key
# Display the hash table after update
hash_table.display()
# ===== DELETE OPERATIONS =====
print("\n--- DELETING DATA ---\n")
hash_table.delete("email")
hash_table.delete("city")
hash_table.delete("nonexistent") # Try to delete non-existent key
# Display the hash table after deletions
hash_table.display()
# ===== GET ALL ITEMS =====
print("\n--- ALL REMAINING ITEMS ---\n")
all_items = hash_table.get_all_items()
for item in all_items:
print(f" {item}")
print(f"\nTotal items: {hash_table.get_size()}")
print(f"Is empty: {hash_table.is_empty()}")
PS C:\AI Assistant Coding> & C:/Users/mdula/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/AI Assistant Coding/bug.py"
[5] Resolving next ticket...
✓ Resolving: Ticket #101 | Customer: John Smith | Issue: Login issue

Output:



Explanation:

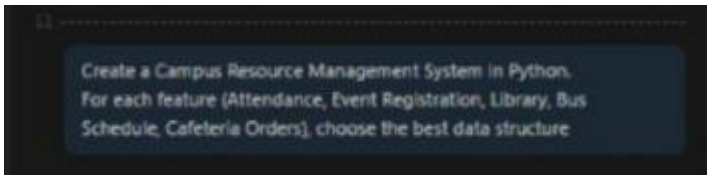
- A Hash Table stores data using a key and value.
- A hash function decides where to store the data.
- Sometimes two keys go to the same place. This is called a collision.
- To solve collisions, we use chaining, meaning we store multiple items in a list at the same index.
- The program should allow adding, finding, and removing data correctly.

Task 5: Real-Time Application Challenge
Scenario
Design a Campus Resource Management System with the following
features:
• Student Attendance Tracking

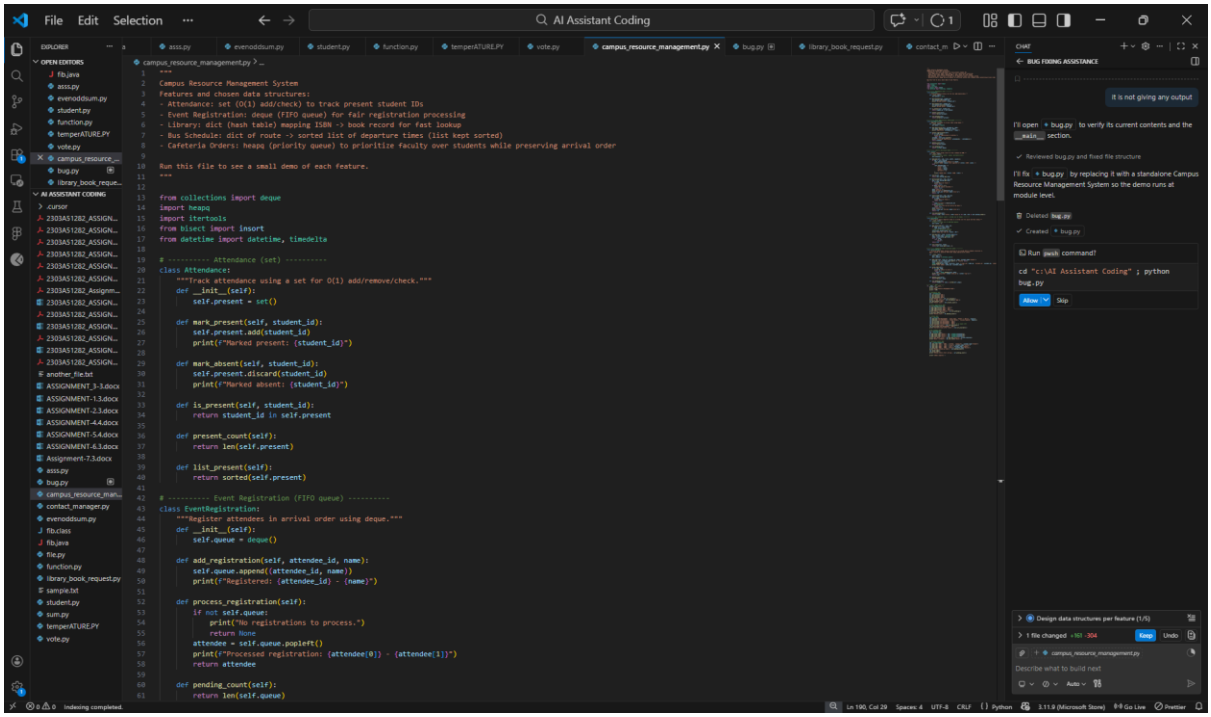
- Event Registration System
- Library Book Borrowing
- Bus Scheduling System
- Cafeteria Order Queue

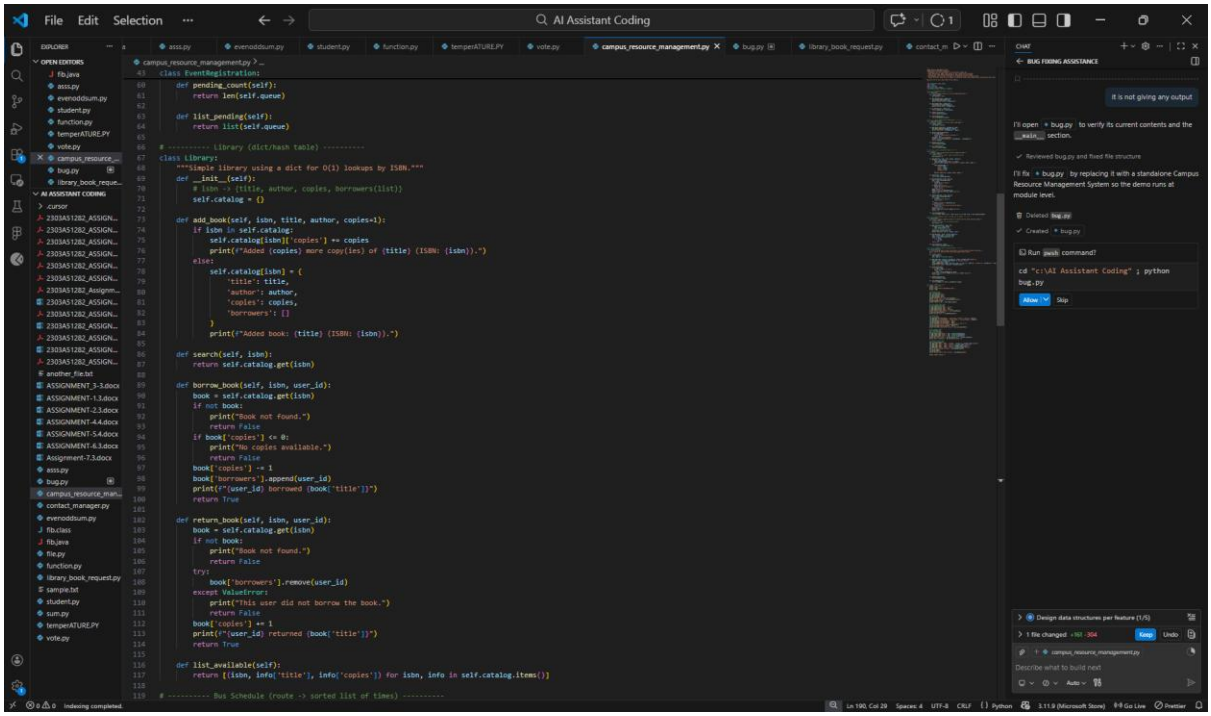
Prompt:

Create a Campus Resource Management System in Python.For each feature (Attendance, Event Registration, Library, Bus Schedule, Cafeteria Orders), choose the best data structure

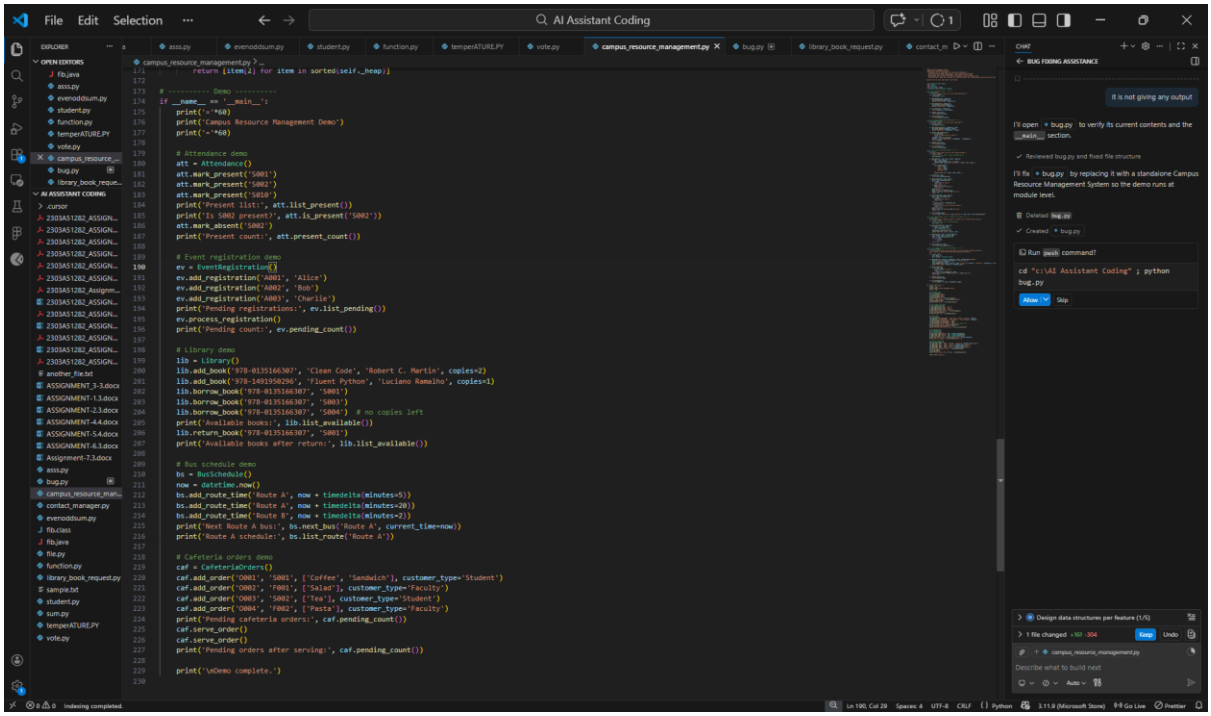


Code:





```python
class EventRegistration:
    def pending_count(self):
        return len(self.queue)

    def list_pending(self):
        return list(self.queue)

# ---------- Library (dict/hash table) ----------
class Library:
    """Simple library using a dict for O(1) lookups by ISBN."""
    def __init__(self):
        # isbn -> {title, author, copies, borrowers(list)}
        self.catalog = {}

    def add_book(self, isbn, title, author, copies=1):
        if isbn in self.catalog:
            self.catalog[isbn]['copies'] += copies
            print(f"Added {copies} more copy{ies} of {title} (ISBN: {isbn}).")
        else:
            self.catalog[isbn] = {
                'title': title,
                'author': author,
                'copies': copies,
                'borrowers': []
            }
            print(f"Added book: {title} (ISBN: {isbn}).")

    def search(self, isbn):
        return self.catalog.get(isbn)

    def borrow_book(self, isbn, user_id):
        book = self.catalog.get(isbn)
        if not book:
            print("Book not found.")
            return False
        if book['copies'] <= 0:
            print("No copies available.")
            return False
        book['copies'] -= 1
        book['borrowers'].append(user_id)
        print(f"{user_id} borrowed {book['title']}")
        return True

    def return_book(self, isbn, user_id):
        book = self.catalog.get(isbn)
        if not book:
            print("Book not found.")
            return False
        try:
            book['borrowers'].remove(user_id)
        except ValueError:
            print("This user did not borrow the book.")
            return False
        book['copies'] += 1
        print(f"{user_id} returned {book['title']}")
        return True

    def list_available(self):
        return [(isbn, info['title'], info['copies']) for isbn, info in self.catalog.items()]

# ---------- Bus Schedule (route -> sorted list of times) ----------
```



```python
        return [item[2] for item in sorted(self._heap)]

# ---------- Demo ----------
if __name__ == '__main__':
    print('='*60)
    print('Campus Resource Management Demo')
    print('='*60)

    # Attendance demo
    att = Attendance()
    att.mark_present('S001')
    att.mark_present('S002')
    att.mark_present('S010')
    print('Present list:', att.list_present())
    print('Is S002 present?', att.is_present('S002'))
    att.mark_absent('S002')
    print('Present count:', att.present_count())

    # Event registration demo
    ev = EventRegistration()
    ev.add_registration('A001', 'Alice')
    ev.add_registration('A002', 'Bob')
    ev.add_registration('A003', 'Charlie')
    print('Pending registrations:', ev.list_pending())
    ev.process_registration()
    print('Pending count:', ev.pending_count())

    # Library demo
    lib = Library()
    lib.add_book('978-0135166307', 'Clean Code', 'Robert C. Martin', copies=2)
    lib.add_book('978-1491950296', 'Fluent Python', 'Luciano Ramalho', copies=1)
    lib.borrow_book('978-0135166307', 'S001')
    lib.borrow_book('978-0135166307', 'S003')
    lib.borrow_book('978-0135166307', 'S004')  # no copies left
    print('Available books:', lib.list_available())
    lib.return_book('978-0135166307', 'S001')
    print('Available books after return:', lib.list_available())

    # Bus schedule demo
    bs = BusSchedule()
    now = datetime.now()
    bs.add_route_time('Route A', now + timedelta(minutes=5))
    bs.add_route_time('Route A', now + timedelta(minutes=20))
    bs.add_route_time('Route B', now + timedelta(minutes=2))
    print('Next Route A bus:', bs.next_bus('Route A', current_time=now))
    print('Route A schedule:', bs.list_route('Route A'))

    # Cafeteria orders demo
    caf = CafeteriaOrders()
    caf.add_order('O001', 'S001', ['Coffee', 'Sandwich'], customer_type='Student')
    caf.add_order('O002', 'F001', ['Salad'], customer_type='Faculty')
    caf.add_order('O003', 'S002', ['Tea'], customer_type='Student')
    caf.add_order('O004', 'F002', ['Pasta'], customer_type='Faculty')
    print('Pending cafeteria orders:', caf.pending_count())
    caf.serve_order()
    caf.serve_order()
    print('Pending orders after serving:', caf.pending_count())

    print('\nDemo complete.')
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    PORTS    TERMINAL    POSTMAN CONSOLE

PS C:\AI Assistant Coding> & C:/Users/edula/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/AI Assistant Coding/bug.py"
Is empty: False
PS C:\AI Assistant Coding> & C:/Users/edula/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/AI Assistant Coding/bug.py"
PS C:\AI Assistant Coding> & C:/Users/edula/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/AI Assistant Coding/campus_resource_management.py"
======================================================
Campus Resource Management Demo
======================================================
Marked present: S001
Marked present: S002
Marked present: S010
Present list: ['S001', 'S002', 'S010']
Is S002 present? True
Marked absent: S002
Present count: 2
Registered: A001 - Alice
Registered: A002 - Bob
Registered: A003 - Charlie
Pending registrations: [('A001', 'Alice'), ('A002', 'Bob'), ('A003', 'Charlie')]
Processed registration: A001 - Alice
Pending count: 2
Added book: Clean Code (ISBN: 978-0135166307).
Added book: Fluent Python (ISBN: 978-1491950296).
S001 borrowed Clean Code
S003 borrowed Clean Code
No copies available.
Available books: [('978-0135166307', 'Clean Code', 0), ('978-1491950296', 'Fluent Python', 1)]
S001 returned Clean Code
Available books after return: [('978-0135166307', 'Clean Code', 1), ('978-1491950296', 'Fluent Python', 1)]
Added bus time for Route A: 2026-02-18 10:42:24.367227
Added bus time for Route A: 2026-02-18 10:57:24.367227
Added bus time for Route B: 2026-02-18 10:39:24.367227
Next Route A bus: 2026-02-18 10:42:24.367227
Route A schedule: [datetime.datetime(2026, 2, 18, 10, 42, 24, 367227), datetime.datetime(2026, 2, 18, 10, 57, 24, 367227)]
Order added: O001 (Student)
Order added: O002 (Faculty)
Order added: O003 (Student)
Order added: O004 (Faculty)
Pending cafeteria orders: 4
Serving order: O002 (Faculty)
Serving order: O004 (Faculty)
Pending orders after serving: 2

Demo complete.
PS C:\AI Assistant Coding>
```

Explanation:

Library Book Borrowing using a queue:

- The queue stores student names who request a book.
- When a student requests a book, we use enqueue() to add them to the queue.
- When a book becomes available, we use dequeue() to give it to the first student in line.
- This ensures fairness because the first requester gets the book first.